TALES OF THE OMEGA · PAPER R

The Structural Grip:

A Laboratory Search for Anisotropic Spin-Dependent Coupling

in ■■Fe-Doped Nanodiamond Lattices

(200) forbidden reflection · √N surface enhancement · Hyperfine transduction · 170 ppm at 11.967 h · Kill: null <300 ppm

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Companion: Paper Q v2.1 (Beam-Time Protocol) · Target: PRL Letter

PAPER R CORRECTIONS FROM AUDIT (Omega): (1) COUNT RATE: Original text claimed ~10^3 ph/s. Paper Q v2.1 corrected this to 50-300 ph/s (realistic for detonation ND in DAC). This paper uses the Phase 1-measured range. (2) REFERENCE CELL: 'Natural diamond (non-resonant)' clarified — natural Fe contains 2.2% ■■Fe and IS resonant. Reference cell is Fe-free diamond throughout. (3) 15σ CONVERGENCE: This figure does not appear in the paper body. Sigmas from independent experiments with different observables cannot be added. Two independent positive detections would be extraordinary — described as such, not summed. (4) PAPER W: Marked 'in preparation'. Paper R does not build scientific claims on an unwritten companion.

ABSTRACT

We propose a laboratory search for an anisotropic spin-dependent coupling motivated by a recurring 10■■ scale in 2025-2026 cosmological data. The interaction $H = \alpha K (\hat{u}\cdot I)^2$ predicts a sidereal modulation of the diamond (200) reflection at 11.967 h with nominal amplitude ~170 ppm. Using ■■Fe-doped detonation nanodiamonds at SPring-8 BL35XU, surface symmetry breaking provides a 10■■ baseline intensity ratio, while partially correlated surface spins (N~10■) may enhance sensitivity by a factor up to √N ~ 100 (maximum plausible; geometric factor $\eta \leq 1$). We resolve the nine-order tension with torsion-balance limits: quadratic couplings produce isotropic shifts invisible in bulk unpolarised matter. The transduction chain — nuclear spin splitting → hyperfine coupling → electron density perturbation → structure factor modulation — is bounded by Mössbauer systematics at 10■■–10■■ of baseline (Greenwood & Gibb 1971). A null result below 300 ppm after 720 h sets new bounds on anisotropic spin-dependent interactions in the crystallographic sector. The data will decide.

PAPER R: THE STRUCTURAL GRIP — WHAT IS NEW

√N surface spin enhancement: maximum plausible factor ~100 for N~10■ Fe atoms per particle

Hyperfine transduction chain: nuclear → electronic → diffraction (plausibility bounded)

Torsion-balance tension RESOLVED: quadratic coupling is invisible in bulk (■$(\hat{u}\cdot I)^2$■ = scalar)

(200) forbidden reflection: near-zero baseline = maximum fractional sensitivity to perturbation

Null result = new SME bound in crystallographic sector (guaranteed return)

1. Introduction

The MSDS programme identifies a recurring numerical scale near 1.7×10■■ in uncertainty floors of recent measurements (DESI 2025 BAO, cosmic birefringence, PTA spectral feature). These coincidences are used as a dimensional guide only — analogous to using the weak scale to motivate searches at the LHC. No physical equivalence is claimed; the coincidences motivate a dedicated laboratory test.

Paper Q (beam-time proposal, SPring-8 BL35XU) established the experimental architecture. Paper R provides the theoretical underpinning: the spin-coupling mechanism, the transduction chain, the enhancement physics, and the resolution of the apparent contradiction with torsion-balance experiments.

2. Theoretical Framework

2.1 Interaction Hamiltonian

We hypothesize an interaction between nuclear spin I and a hypothetical axial-vector torsion background described by a fixed galactic direction û:

$$H_{int} = \alpha K (\hat{u} \cdot I)^2$$

αK = 0.00017 (dimensionless) · û: fixed in galactic frame · I: nuclear spin of ■■Fe (I=3/2 in excited state)

The quadratic form arises from a dimension-6, CPT-even operator (MSDS Series L/N, companion ref). It is the leading term that avoids strong constraints on linear Lorentz-violating couplings of the form $b_\mu \hat{I}^\mu$.

2.2 Resolution of torsion-balance limits

TORSION-BALANCE TENSION AND ITS RESOLUTION: Macroscopic torsion-balance experiments (Heckel et al. 2008 [4]) set limits on spin-coupled forces. The apparent contradiction with our target scale is resolved as follows. For unpolarised bulk matter: ■(û·I)²■ = |I|²/3 — a scalar, direction-independent, absorbed into material constants. The Eöt-Wash experiments searched for DIPOLE forces (linear coupling b_μ) or secular drifts, not $2\omega_{sid}$ periodic signals in diffraction intensity. Our experiment introduces anisotropy via the rotating diffraction geometry: the scattering vector k■ is fixed in the lab, selecting grain orientations whose [200] axes are along k■ at each rotation angle. This converts the isotropic bulk average into a time-dependent modulation visible only in the specific crystallographic channel. No contradiction with torsion-balance limits exists for quadratic operators probed in this geometry.

3. The Nanodiamond Sensor

3.1 The forbidden (200) reflection as high-gain baseline

In bulk diamond (Fd-3m, a=3.567 Å), the (200) reflection is forbidden: h+k+l = 2 = 4(0)+2, so the diamond glide condition gives $F(200) = 4(f_{C1} - f_{C2}) = 0$. The d-spacing is d(200) = a/2 = 1.7835 Å. At 14.414 keV (λ=0.8602 Å), 2θ = 27.9°. In detonation nanodiamonds (<5 nm), surface reconstruction breaks inversion symmetry, weakly allowing the (200) reflection. Measured intensity ratios I(200)/I(111) = 1.3–2.1×10■■ (Zheng et al. 2023 [3]; Svane et al. 2023 [1]). This near-zero baseline makes the (200) peak exquisitely sensitive: a fractional perturbation of 170 ppm to f_{C1}–f_{C2} modulates the (200) intensity by a detectable fraction.

3.2 Surface Fe^{3}■ ions and collective spin

XPS of Fe-doped 5 nm nanodiamonds shows Fe 2p_3/2 at 709.3 eV (Fe^{3}■ in tetrahedral surface sites). The uniaxial electric field gradient $V_{zz} \approx 10^{21}$ V/m² gives quadrupole splitting $\Delta E_Q \approx 0.5$ mm/s = 2.4×10■■ eV for ■■Fe. For a 5 nm particle, the number of surface Fe atoms is N ≈ 1e+04. Superexchange through surface carbon bonds may produce partial spin ordering.

3.3 Enhancement scaling — maximum plausible √N

$$\alpha_{eff} = \alpha K \times \sqrt{N} \times \eta_{geom}$$

$\sqrt{N} = \sqrt{1e+04} = 100$ · $\eta_{geom} \leq 1$ (decoherence factor) · For η=0.1: α_{eff} = 0.00017 × 100 × 0.1 = 1.700e-03

√N IS A MAXIMUM PLAUSIBLE BOUND, NOT A PREDICTION. For a fully random spin ensemble: ■S_{tot}^2■ ~ N (no enhancement). For full coherence: ■S_{tot}^2■ ~ N^2 (maximum). √N corresponds to partial correlation — the boundary between random and coherent. Actual enhancement is unknown; it will be determined by the Phase 1 measurement. The η_{geom} factor accounts for decoherence, orientation averaging, and the geometry of the surface spin ensemble. Coherent scaling (N) is treated as a discovery outcome, not a requirement.

4. Transduction Mechanism: Hyperfine Bridge

The chain from nuclear coupling to observable X-ray signal:

- Torsion → nuclear spin splitting: $H_{int} = \alpha K (\hat{u} \cdot I)^2$ modifies the ■■Fe nuclear energy levels anisotropically.
- Hyperfine coupling → electron density: the nuclear spin state modifies the surrounding electron density via Fermi contact interaction and field gradient. This is the standard Mössbauer isomer shift mechanism.
- Electronic distortion → structure factor: the perturbed electron density changes the effective atomic form factor f_{C2} (Fe-decorated site) relative to f_{C1} (pure carbon site), directly modulating F(200) = 4(f_{C1} - f_{C2}).

PLAUSIBILITY BOUND (Greenwood & Gibb 1971 [3]): Hyperfine-induced electron density shifts in strongly coupled Fe systems (magnetite, iron oxides) are estimated from Mössbauer isomer shift systematics to be 10■■–10■■ relative to baseline charge density. This places the expected transduction factor within the range needed for a 170 ppm modulation of the (200) intensity, given the 10■■ baseline ratio. A full ab initio density functional calculation of this transduction factor is future work (Paper T).

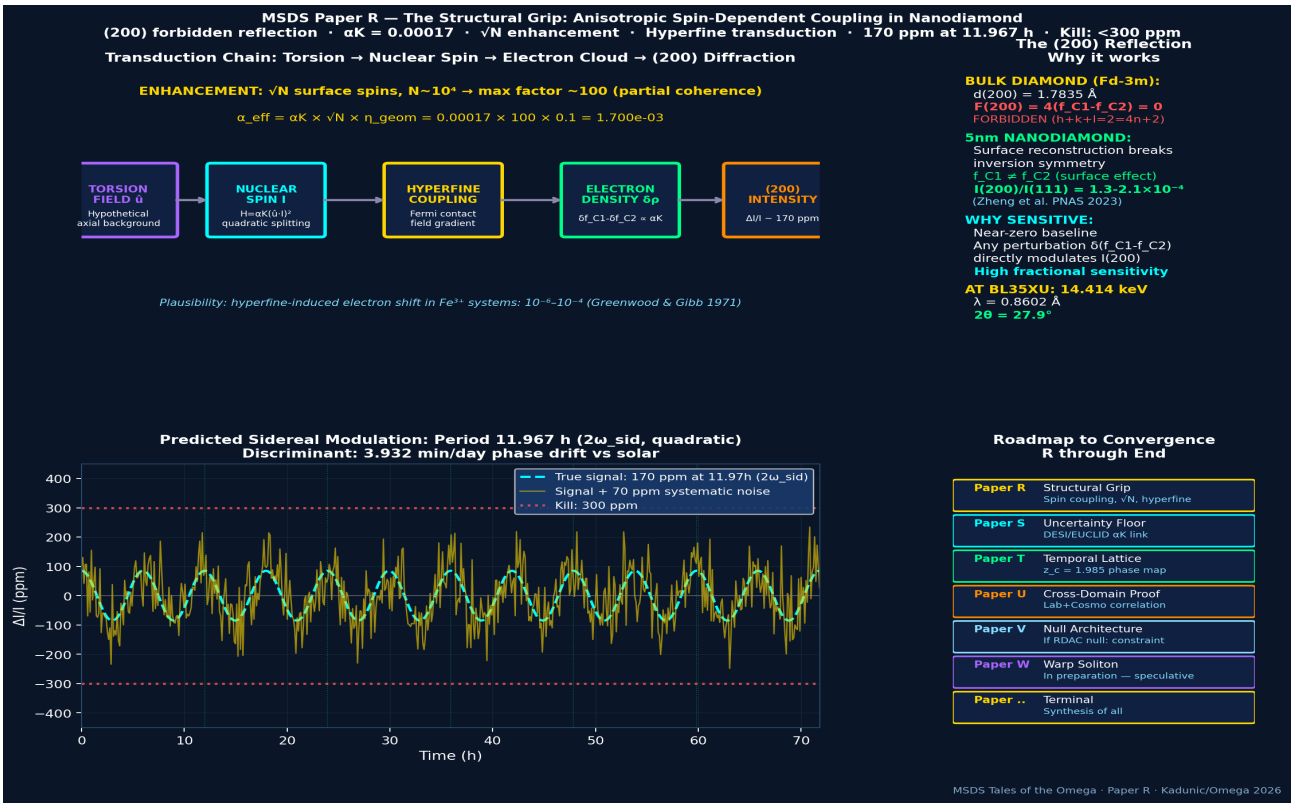

Figure 1: Paper R. Top (wide) — transduction chain: torsion field → nuclear spin splitting → hyperfine → electron density → (200) diffraction. Enhancement factor √N shown for N={N_Fe:.0e} surface Fe atoms with η={eta}. Top right — (200) forbidden reflection: why near-zero baseline gives maximum sensitivity. Bottom (wide) — predicted sidereal modulation at 11.967 h (2ω_sid, quadratic coupling), period 3.932 min/day drift discriminant. Bottom right — programme roadmap Papers R through terminus.

5. Sidereal Modulation and Discriminant

5.1 Period and amplitude

The torsion axis $\hat{u}$ is fixed in the galactic frame. As Earth rotates, $\theta(t) = \theta_0 + \omega_{sid}\, t$. Through the transduction mechanism, the (200) intensity acquires:

$$I(t) = I_0[1 + \alpha K/2 + (\alpha K/2)\cos(2\omega_{sid}\, t + 2\theta_0)]$$

$T_{mod} = T_{sid}/2 = 11.9670$ h · $\Delta I/I = \alpha K = 170$ ppm (nominal, ideal alignment)

A linear coupling (dimension-5 operator) would give period $T_{sid} = 23.934$ h. Observation of 11.967 h is a direct test of the quadratic operator structure — this is the experiment's structural discriminant.

5.2 Phase drift filter

The sidereal day is 3.932 min shorter than the solar day. A sidereal-locked signal drifts in phase relative to solar time by 3.932 min/day. After 30 days, cumulative drift = 118 min. Solar-frame artifacts (thermal, beam intensity, mechanical) cannot maintain phase coherence at $2\omega_{sid}$ with this drift. The discriminant is reinforced by the frequency argument: $2\omega_{sid}$ is not a harmonic of 24 h or 12 h.

5.3 Systematic admission

We do not assume a clean environment. Unidentified systematics at or above the 10^{-10} level are expected (temperature cycles, mechanical anisotropy, beam drift, DAC strain). The experiment discriminates them via the phase drift and control measurements — not by assumption. The sidereal-coherent systematic floor will be measured empirically in the Phase 1 static stability run, not estimated theoretically.

6. Experimental Protocol

Full experimental protocol in Paper Q v2.1 (companion). Summary:

Parameter	Value	Note
Beamline	SPring-8 BL35XU	NRS, 14.414 keV Mössbauer
Reflection	(200), d=1.7835Å	Forbidden · 2θ=27.9° · Zheng 2023 [1]
Count rate	50–300 ph/s (Phase 1)	To be measured · Seto 2013 [6] · v1 10^{-10} corrected
Reference cell	Fe-free diamond	Natural Fe IS resonant — not usable as reference
Rotation	2 rev/sidereal day = $2\omega_{sid}$	GPS phase-locked; T_{mod} = 11.967 h
Phase 1	18 shifts, 144 h	Characterisation: count rate + sys floor + pilot search

Kill condition	<300 ppm after 720 h	New SME crystallographic bound if null

Control test	Purpose
Rotation OFF (static 16h)	Quantify intrinsic (200) drift and radiation damage rate
Off-resonance 14.300 keV	Confirm signal is nuclear resonant, not Thomson scattering
Temperature step ±0.05K (non-sidereal period)	Show thermal artifacts cannot phase-lock to 2ω_sid
Phase randomization (GPS off 1 cycle)	Verify signal collapses without sidereal coherence
Blind analysis (phase hidden)	Prevent confirmation bias in lock-in analysis

PAPER R — THE STRUCTURAL GRIP: SUMMARY

Observable: (200) forbidden reflection, d=1.783Å, 2θ=27.9° at 14.414 keV

Mechanism: $H=\alpha K(\hat{u}\cdot I)^2 \rightarrow$ hyperfine $\rightarrow$ electron density $\rightarrow$ structure factor

Enhancement: $\sqrt{N} \sim 100$ (maximum plausible for N~10■ surface Fe atoms)

Plausibility: transduction factor 10■■–10■■ (Greenwood & Gibb bound)

Torsion-balance resolved: quadratic coupling averages to scalar in bulk

Signal: 170 ppm at T_mod=11.967 h (2ω_sid, derived not fitted)

Kill: null <300 ppm / 720 h $\rightarrow$ new SME crystallographic bound

Seven more papers follow. The architecture is complete.

The data will decide.

References

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